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Cauchy sequences

Problem 25

(a) Show that if $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence, then $\lim_{n\to\infty}(a_{n+1} - a_n) = 0$.

(b) Use the definition of a Cauchy sequence to show that the sequence of partial sums of the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is not a Cauchy sequence.

(c) Give an example of a sequence $(a_n)_{n=1}^{\infty}$ with $\lim_{n\to\infty}(a_{n+1} - a_n) = 0$ which is not a Cauchy sequence.

Absolute and conditional convergence

Problem 26 For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent.

(a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{2n+1}}$, (b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+1)}$, (c) $\sum_{n=2}^{\infty} (-1)^n \frac{n}{\ln n}$,

(d) $\sum_{n=0}^{\infty} (-1)^n \frac{5}{n^3+1}$, (e) $\sum_{n=0}^{\infty} \frac{(-5)^n}{n!}$, (f) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^2+3}{(2n-5)^2}$,

(g) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt[3]{n}}{n+1}$, (h) $\sum_{n=1}^{\infty} \frac{\cos\left(n\frac{\pi}{6}\right)}{n^2}$, (i) $\sum_{n=1}^{\infty} \frac{n^n}{(-3)^n}$.

Alternating series

Problem 27 For each of the following series, determine how many terms of the series are required to approximate the sum of the series to four decimal places. The latter means that the difference between the sum and the partial sum must be $< \frac{1}{2} \cdot 10^{-4}$.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$, (b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^n}$.

The root test

Problem 28 Prove the Root Test:

Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers such that the limit

$$L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}}$$

exists. Then the following holds.

(1) If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(2) If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

(3) If $L = 1$, then the test is inconclusive.